

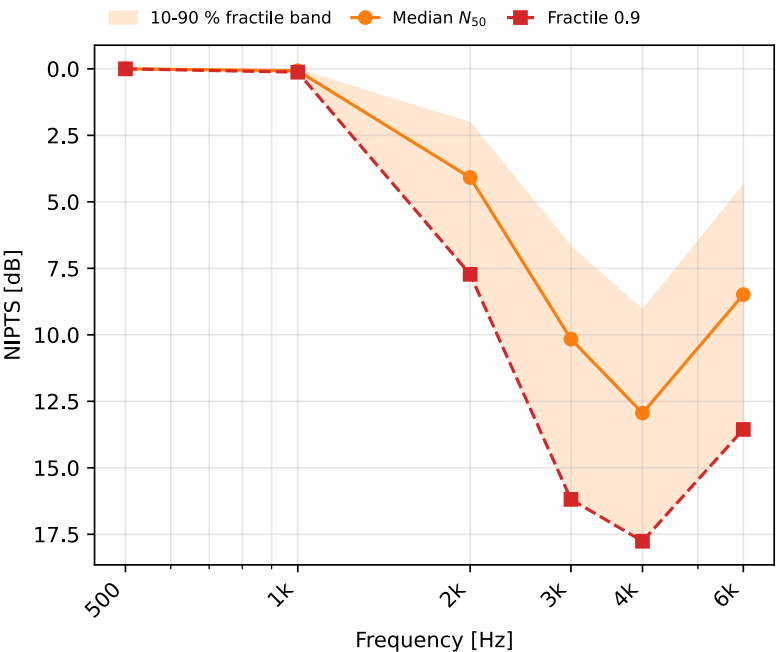
# Noise-induced hearing loss prediction

Statistical prediction of the noise-induced permanent threshold shift of a noise-exposed population per ISO 1999:2013 (clause 6.3).

|            |                             |                     |                                                 |
|------------|-----------------------------|---------------------|-------------------------------------------------|
| Company:   | Example fabrication works   | Worker(s) / group:  | Welders (homogeneous exposure group, 4 workers) |
| Workplace: | Steel assembly hall, line 2 | Date of assessment: | 2026-07-20                                      |

Threshold shift by frequency

| Frequency [Hz] | N <sub>50</sub> [dB] | NIPTS [dB] |
|----------------|----------------------|------------|
| 500            | 0.0                  | 0.0        |
| 1000           | 0.1                  | 0.1        |
| 2000           | 4.1                  | 7.7        |
| 3000           | 10.2                 | 16.2       |
| 4000           | 12.9                 | 17.8       |
| 6000           | 8.5                  | 13.6       |



**Predicted NIPTS averaged over 2/3/4 kHz = 13.9 dB**

Noise exposure  $L_{EX,8h} = 90$  dB  
Exposure duration = 20 years  
Population fractile  $Q = 10$  % (fraction with worse hearing)

These values are a statistical prediction for a noise-exposed population (ISO 1999:2013), not a clinical diagnosis or a measured audiogram of any individual.

$Q$  is the percentage of the noise-exposed population predicted to show a larger (worse) value than the one stated, as ISO 1999:2013 defines it (6.3.2, Formulae (4) and (5));  $Q = 10$  % is the most-susceptible tenth.

ISO 1999:2013 does not specify frequencies or frequency combinations for evaluating hearing disability (Scope, NOTE 1); the 2/3/4 kHz average shown is a commonly used descriptor whose choice is left to the user.

**Laboratory:** Phonometry reference example

**Report no.:** EXAMPLE-1999-NIPTS

**Date:** 2026-07-20

**Notes:** Reproduces the ISO 1999:2013 Annex D (Table D.2) worked example.

Operator: phonometry    Signature: \_\_\_\_\_

■ The results are a statistical prediction for the stated population and exposure conditions; they do not describe any individual person.  
Generated by phonometry.